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SYSTEM FOR CONTROLLING PRODUCTION OF OZONE FOR AN INTERNAL COMBUSTION ENGINE

Abstract

A method of using ozone for maintaining an internal combustion engine. The method includes providing a maintenance system including a user control panel, an ozone generator, a blower, the blower having a first blower end for intake of air and a second blower end connected to the ozone generator and a safety module with at least one safety sensor connected thereto. The maintenance system includes a control module connected to the ozone generator and to the safety module. The system includes an ozone transport tube having a first tube end sealable to the internal combustion engine. The method include connecting the ozone transport tube to an air intake of the internal combustion engine and directing the ozone through the ozone transport tube during operation of the internal combustion engine. The method includes allowing the ozone to react with deposits in the internal combustion engine.

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Background/Summary

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REFERENCE TO RELATED APPLICATIONS

[0003] This application is a Non-Provisional Utility Patent application claiming priority to U.S. patent application Ser. No. 18/587,001, filed on Feb. 26, 2024, the application of which is incorporated by reference herein in its entirety.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0004] The invention relates in general to the field of internal combustion engine maintenance, and more particularly to a system which produces and controls ozone for removal of deposits injected into the fuel or air path of an internal combustion engine, which is used to improve the performance efficiency and emissions of that internal combustion engine.

DESCRIPTION OF RELATED ART

[0005] Since the development of the internal combustion engine there has always been a desire to improve performance by improving the miles per gallon achieved when such engines are used in moving vehicles such as trucks, buses and automobiles. More recently there has been a desire to maintain a more complete combustion in the internal combustion engine, especially in a diesel fuel engine, and eliminate unhealthy emissions such as non-combusted fuel and fuel particles, carbon monoxide, nitrogen oxides, and ozone.

[0006] Engine carbon deposits have a measurable effect on performance, emissions, and fuel economy. Carbon deposits have been a problem for the internal combustion engine since it was invented. It is well known that carbon deposits in internal combustion engines reduce engine performance, reduce fuel efficiency, shorten engine/equipment/parts life, and increase engine out emissions.

[0007] There is a need for a system to maintain low carbon deposits during operational activities that include constant low rev driving, short journeys, continual stop-start journeys, long idling, and poor fuel quality.

SUMMARY OF THE INVENTION

[0008] The instant invention in one form is directed to a system for controlling production of ozone for an internal combustion engine. The system includes an ozone generator mounted to an engine of a vehicle. The ozone generator includes a tubular containment shell having a first and second opening for allowing air to flow therethrough, and first and second cylindrical electrodes disposed inside the tubular containment shell permitting air to flow through. The system includes a high voltage source connected between the first and second electrode, the high voltage source capable of producing a voltage differential between the first and second electrodes sufficient to cause the production of ozone. The ozone generator is disposable in an air intake system of a vehicle having the internal combustion engine, the ozone generator permitting air to flow therethrough wherein a

change from oxygen to ozone occurs in the air flowing through the ozone generator and into the internal combustion engine. The system includes at least one control module for controlling production of ozone by the ozone generator. The control module communicates with a power inverter in the ozone generator, a sensor on the ozone generator, and the internal combustion engine. The control module may be a frequency control module or a voltage control module for controlling the ozone generator voltage for maximizing performance. The control module may be a temperature control module for activating a cooling system of the ozone generator or the control circuit. The control module may be a flow control module for adjusting air flow through the ozone generator. The control module may be a timer control module for determining state of ozone generator (ON/OFF) as a function of mileage and time and for keeping track of mileage and time. The control module may be an emergency shutoff module.

[0009] The system may include at least one monitor module. The monitor module communicates with the control module, the sensor on the ozone generator, and the internal combustion engine. The monitor module includes monitoring features which may be incorporated into the at least one control module. The monitor module is for detecting engine parameters and for monitoring the production of ozone. The monitor module may be a frequency monitor module or a voltage monitor module having a voltage sensor disposed on the ozone generator for determining the voltage supplied to the ozone generator. The monitor module may be a temperature monitor module including at least one temperature sensor for determining the temperature of the ozone generator, control circuit, or a portion of the internal combustion engine. The monitor module may be an air flow monitor which monitors air flow through the ozone generator. The monitor module may be a timer monitor module for generating time interval for the control circuit to work with. The monitor module may be an emergency monitor module for alerting a user to an engine or generation system having an out of range parameter. The monitor module monitors structural parameters including the integrity of each component of the ozone generator. The monitor module monitors fuel parameters including instant fuel usage, average fuel usage, fuel put into the fuel storage tank and other static or dynamic fuel functions. The monitor module may be integrated with the at least one control module to perform both monitoring and controlling functions.

[0010] The instant invention in another form is directed to an ozone generation system for an internal combustion engine. The ozone generation system includes an ozone generator having a conductive cylindrical containment shell having a first and second opening for allowing air to flow therethrough. The system includes a first cylindrical electrode and a second cylindrical electrode, both disposed inside the conductive cylindrical containment. The system includes a high voltage source connected between the first and second cylindrical electrode, the high voltage source capable of producing a voltage differential between the first and second cylindrical electrode sufficient to cause production of ozone. The ozone generator permits air to flow therethrough wherein a change from oxygen to ozone occurs in the air flowing through the ozone generator. The system includes at least one monitor module including a voltage sensor disposed on the ozone generator for sensing the voltage differential between the first and second cylindrical electrode and at least one control module for controlling the voltage differential between the first and second cylindrical electrode. The system may include a dielectric element disposed between the first and second cylindrical electrode. The dielectric element is preferably glass and more preferably quartz glass. The system may include a temperature monitor module including at least one temperature sensor. The system may include an air flow monitor for monitoring air flow through the ozone generator. The system may include an emergency monitor module for alerting a user to an engine or generation system having an out of range parameter. The system may include an emergency shutoff module for terminating power to the high voltage source. The system may include an engine monitor module for detecting engine parameters.

[0011] In another aspect, the system includes an ozone generator having a tubular containment shell having a first and second opening for allowing air to flow therethrough, a first and second

cylindrical electrode disposed inside the tubular containment shell permitting air to flow therethrough and a high voltage source connected between the first and second electrode, the high voltage source capable of producing a voltage differential between the first and second electrode sufficient to cause production of ozone. The ozone generator is disposable in an air intake system of a vehicle having the internal combustion engine. The ozone generator permits air to flow therethrough wherein a change from oxygen to ozone occurs in the air flowing through the ozone generator and into the internal combustion engine. The system includes at least one monitor module for monitoring parameters of the system and at least one control module for controlling the production of the ozone. The monitor module may measure a first temperature of the ozone generator and a second temperature of the high voltage source. The ozone generator may include a dielectric element disposed between the first and second electrode. The dielectric element may be glass. The tubular containment shell may be cylindrical. The first and second electrode may be coaxially aligned with the tubular containment shell, the first electrode spaced a distance from the second electrode, the first electrode being aligned axially within the second electrode. The control module is a frequency control module for controlling modulation frequency of the high voltage source. The control module is a module for adjusting air flow through the ozone generator. The control module may also be a timer control module for maintaining records of time and vehicle mileage. The control module may be an emergency shutoff module for disconnecting the high voltage source from the first and second electrode. The monitor module monitors structural parameters including integrity of each component of the ozone generator.

[0012] In another aspect, the system includes an ozone generator including a conductive cylindrical containment shell having a first and second opening for allowing air to flow therethrough, first and second cylindrical electrodes disposed inside the conductive cylindrical containment shell permitting air to flow therethrough, and a high voltage source connected between the first and second electrode, the high voltage source capable of producing a voltage differential between the first and second electrodes sufficient to cause production of ozone. The ozone generator permits air to flow therethrough wherein a change from oxygen to ozone occurs in the air flowing through the ozone generator. The system includes at least one control module selected from a first group consisting of a frequency control module, a voltage control module for controlling an ozone generator voltage for maximizing performance, an air flow control module for adjusting air flow through the ozone generator, a timer control module for determining state of ozone generator (ON/OFF) as a function of mileage and time, and for keeping track of mileage and time, and an emergency shutoff module. The system includes at least one monitor module selected from a second group of a frequency monitor module, a voltage monitor module including a voltage sensor disposed on the ozone generator for determining the voltage differential supplied to the ozone generator, a temperature monitor module including at least one temperature sensor for determining a temperature of a portion of the system, an air flow monitor for monitoring the air flow through the ozone generator, a timer monitor module for generating time intervals for the control module and the monitor module, and an emergency monitor module for alerting a user to the engine or the system having an out of range parameter.

[0013] In another aspect, a system for using ozone for an internal combustion engine includes an ozone generator. The ozone generator includes a containment shell having a first and second opening for allowing air to flow therethrough, first and second concentric cylindrical electrodes disposed inside the containment shell and a high voltage source connected between the first and second concentric cylindrical electrode, the high voltage source capable of producing a voltage differential between the first and second concentric cylindrical electrode sufficient to cause production of ozone. The ozone generator permits air to flow therethrough wherein a change from oxygen to ozone occurs in the air flowing through the ozone generator. The system includes a safety module including at least one ozone sensor for sensing the ozone exterior to the containment shell and a control module for controlling the voltage differential from the high voltage source. The

control module includes an input from the at least one ozone sensor. Controlling the voltage applied between the first and second concentric cylindrical electrode may include controlling amplitude of the voltage or controlling frequency of the voltage or controlling the amplitude and frequency of the voltage.

[0014] In another aspect, a system uses ozone for cleaning of an internal combustion engine. The system includes a case having a case housing and an inner cavity. The case includes a user control panel having a plurality of user controls, the user control panel disposed on the case housing and an ozone generator in the inner cavity, the ozone generator including a containment shell having a first and second opening for allowing air to flow therethrough, a first and second concentric cylindrical electrode disposed inside the containment shell and a controllable high voltage source connected between the first and second cylindrical electrode wherein the containment shell permits air to flow therethrough wherein a change from oxygen to ozone occurs in the air flowing through the containment shell. The case includes a blower having a first blower end for intake of air external to the case and a second blower end connected to the containment shell first opening and a safety module including at least one safety sensor connected thereto, a control module connected to the user control panel and the safety module, the control module for controlling voltage applied between the first and second concentric cylindrical electrode. The system includes a tube connector having a first connector end connected to the containment shell second opening and a second connector end disposed on the case housing. The system includes an ozone transport tube having a first tube end scalable to the second tube connector end and a second tube end for connection to the internal combustion engine. The blower directs the ozone through the tube ozone generator, the tube connector and the ozone transport tube during operation of the internal combustion engine, whereby the ozone reacts with deposits in the internal combustion engine for removal of the deposits. Controlling the voltage applied between the first and second concentric cylindrical electrode may include controlling amplitude of the voltage or controlling frequency of the voltage or controlling the amplitude and frequency of the voltage. The system may include a dielectric element disposed between the first and second cylindrical electrode. The system may include an engine monitor module for detecting engine parameters.

[0015] In another aspect, a method of using ozone for maintaining an internal combustion engine, the method includes providing a maintenance system having a case with a case housing and an inner cavity. The case includes a user control panel inside the inner cavity, an ozone generator inside the inner cavity, a blower inside the inner cavity, the blower having a first blower end for intake of air external to the case and a second blower end connected to the ozone generator and a safety module including at least one safety sensor connected thereto. The maintenance system includes a control module connected to the ozone generator and the safety module, the control module for controlling a voltage applied to the ozone generator, a tube connector having a first connector end connected to the ozone generator and a second connector end disposed on the case housing and an ozone transport tube having a first tube end scalable to the second tube connector end and a second tube end for connection to the internal combustion engine. The method include connecting the second tube end of the ozone transport tube to an air intake of the internal combustion engine, ensuring the internal combustion engine is running and commencing production of the ozone and directing the ozone through the ozone transport tube during operation of the internal combustion engine. The method includes allowing the ozone to react with deposits in the internal combustion engine and after a specific amount of time, stopping production of the ozone and disconnecting the second tube end of the ozone transport tube from the air intake of the internal combustion engine. The control module may adjust air flow through the ozone generator. The control module may include a timer control for determining an amount of time ozone is generated by the ozone generator. The control module may include an emergency shutoff for terminating production of the ozone. The control module may monitor structural parameters including integrity of each component of the ozone generator.

[0016] In another aspect, a system uses ozone for an internal combustion engine and includes an ozone generator. The ozone generator includes a containment shell having a first and second opening for allowing air to flow therethrough, first and second concentric cylindrical electrodes disposed inside the containment shell and a high voltage source connected between the first and second concentric cylindrical electrode, the high voltage source capable of producing a voltage differential between the first and second concentric cylindrical electrode sufficient to cause production of ozone. The ozone generator permits air to flow therethrough wherein a change from oxygen to ozone occurs in the air flowing through the ozone generator. The system includes a safety module having at least one sensor for sensing the ozone exterior to the containment shell or high voltage exterior to the containment shell. The system includes a control module for controlling the voltage differential from the high voltage source. The control module includes an input from the at least one sensor. There is preferably at least one ozone sensor in the system. Controlling the voltage includes controlling amplitude of the voltage or controlling frequency of the voltage or a combination of controlling the frequency and the voltage.

[0017] In another aspect, the invention is a method of using ozone for maintaining an internal combustion engine using the system described above. The method includes connecting the containment shell to an air intake of the internal combustion engine, ensuring the internal combustion engine is running, commencing production of the ozone and directing ozone into the air intake of the internal combustion engine, allowing the ozone to react with deposits in the internal combustion engine. The method may include providing timing intervals of ozone production based on engine performance feedback. The method may include providing timing intervals of ozone production based on sensor feedback. The method may include providing timing intervals of ozone production based on AI feedback through a AI database system trained on vehicle manufacturing data and system feedback.

[0018] These and other objects, features, and advantages of the present invention will become more readily apparent from the attached drawings and the detailed description of the preferred embodiments, which follow.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] A further understanding of the nature and advantages of particular embodiments may be realized by reference to the remaining portions of the specification and the drawings, in which like reference numerals are used to refer to similar components. When reference is made to a reference numeral without specification to an existing sub-label, it is intended to refer to all such multiple similar components.

[0020] FIG. 1 is a block diagram of the ozone generation system for an internal combustion engine according to at least one embodiment of the present invention;

[0021] FIG. 2 is a block diagram of the ozone generation system for a diesel engine according to at least one embodiment of the present invention;

[0022] FIG. 3 is a perspective view of the ozone generator according to at least one embodiment of the present invention;

[0023] FIG. 4 is a front elevational view of the ozone generator shown in FIG. 3;

[0024] FIG. 5 is an enlarged view of a portion of the ozone generator shown in FIG. 4;

[0025] FIG. 6 is a rear elevational view of the ozone generator shown in FIG. 3;

[0026] FIG. 7A is a block diagram of a second embodiment of an ozone generation system for a diesel engine according to the present invention;

[0027] FIG. 7B is a block diagram of a third embodiment of an ozone generation system for a diesel engine according to the present invention;

[0028] FIG. **8** is a block diagram of a high voltage circuit for supplying power to the ozone generator;

[0029] FIG. **9** is a block diagram of another embodiment of a high voltage circuit for supplying power to the ozone generator according to the present invention;

[0030] FIG. **10** is a schematic diagram of the components for the circuit shown in FIG. **9**;

[0031] FIG. **11** is a block diagram showing the system for controlling production of ozone for an internal combustion engine;

[0032] FIG. **12** is a block diagram showing an alternate system for controlling production of ozone for an internal combustion engine;

[0033] FIG. **13** is a diagram showing sensors for sensing ozone outside of the containment shell;

[0034] FIG. **14** is a flowchart showing a method for maintaining an internal combustion engine;

[0035] FIG. **15** shows a breakdown of two steps shown in the flowchart of FIG. **14**;

[0036] FIG. **16** shows a further breakdown of steps shown in the flowchart of FIG. **15**;

[0037] FIG. **17** shows a further breakdown of steps shown in the flowchart of FIG. **15**;

[0038] FIG. **18** is a flowchart showing a method of using the system; and

[0039] FIG. **19** is a diagram showing the system for maintaining an internal combustion engine.

[0040] Corresponding reference characters indicate corresponding parts throughout the several views. The exemplifications set out herein illustrate embodiments of the invention and such exemplifications are not to be construed as limiting the scope of the invention in any manner.

DESCRIPTION OF THE PREFERRED EMBODIMENT(S)

[0041] While various aspects and features of certain embodiments have been summarized above, the following detailed description illustrates a few exemplary embodiments in further detail to enable one skilled in the art to practice such embodiments. The described examples are provided for illustrative purposes and are not intended to limit the scope of the invention.

[0042] In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the described embodiments. It will be apparent to one skilled in the art however that other embodiments of the present invention may be practiced without some of these specific details. Several embodiments are described herein, and while various features are ascribed to different embodiments, it should be appreciated that the features described with respect to one embodiment may be incorporated with other embodiments as well. By the same token however, no single feature or features of any described embodiment should be considered essential to every embodiment of the invention, as other embodiments of the invention may omit such features.

[0043] In this application the use of the singular includes the plural unless specifically stated otherwise and use of the terms “and” and “or” is equivalent to “and/or,” also referred to as “non-exclusive or” unless otherwise indicated. Moreover, the use of the term “including,” as well as other forms, such as “includes” and “included,” should be considered non-exclusive. Also, terms such as “element” or “component” encompass both elements and components including one unit and elements and components that include more than one unit, unless specifically stated otherwise.

[0044] Lastly, the terms “or” and “and/or” as used herein are to be interpreted as inclusive or meaning any one or any combination. Therefore, “A, B, or C” mean A; B; or C; and “A, B and/or C” mean “any of the following: A; B; C; A and B; A and C; B and C; or A, B and C.” An exception to this definition will occur only when a combination of elements, functions, steps or acts are in some way inherently mutually exclusive.

[0045] As this invention is susceptible to embodiments of many different forms, it is intended that the present disclosure be considered as an example of the principles of the invention and not intended to limit the invention to the specific embodiments shown and described.

[0046] The term ozone generator is the portion of the ozone generation system which produces the oxygen to ozone reaction.

[0047] The term tubular is meant an elongated hollow structure having tube walls which may be

cylindrical, square, oval or other shape which may be symmetrical or asymmetrical. One example of a tube would be a square tube for shipping or for fabricating metal frame structures.

[0048] Referring now to the drawings FIGS. **1-12**, and more particularly to FIG. **1**, there is shown an embodiment of the ozone generation system for an internal combustion engine according to the present invention.

[0049] The system uses the ozone generator for increasing fuel efficiency and reducing emissions in an internal combustion engine. The system includes an air filter **40** which filters incoming atmospheric air and feeds it to an ozone generator **10**. Ozone generator **10** converts the oxygen in the air to ozone, which is then fed into the internal combustion engine **44** for combustion with a desired fuel, such as diesel, gasoline, ethanol, natural gas, or other liquid or gaseous fuel. The ozone is a stronger oxidation gas than oxygen and allows the fuel to burn more completely, increasing efficiency and passing fewer emissions through the exhaust outlet **46**.

[0050] FIG. **2** shows the block diagram for a system more specifically for a diesel engine and includes an air filter **90** flowing air into the ozone generator **10** which is powered by a high voltage power supply **18**. The flow of air is compressed by a turbocharger fan **96** which pulls air from the ozone generator **10** and air filter **90** and feeds the compressed air to the diesel engine **98**. After the diesel engine burns the fuel, the product of combustion is forced through the exhaust outlet **46**.

[0051] FIGS. **3-6** show a first embodiment of the ozone generator according to the present invention. The containment shell **50** or housing is a rigid tube, which includes first and second openings **32**, **34** at the opposite ends that allow air to pass through the tube when there is a pressure differential between the first opening **32** and the second opening **34**. A cylindrical outer electrode **54** and a cylindrical inner electrode **56** are radially separated by a cylindrical dielectric sleeve **52**, and are coaxially disposed inside the conductive containment shell **50**. The electrodes **54**, **56** are preferably a mesh screen. The dielectric sleeve **52** is preferably a quartz glass although other glass materials or polycarbonate such as Lexan may be used, all of which have high strength dielectric properties. Other dielectric materials may be used. At least one, and preferably three, short supporting insulator members **58** are attached to and extend inwardly between the conductive containment shell **50** and the dielectric sleeve **52** at spaced intervals adjacent each opening **32**. Alternately, the insulator members **58** may be attached to and extend between the conductive tube **50** and at least one of the inner electrode **56**, outer electrode **54** and dielectric sleeve **52**. The insulator members **58** are preferably of polyetheretherketone (PEEK) material although Teflon® or other aggressive environment material may be used. A high voltage insulated wire **80** connects the high voltage source to the inner electrode **56** and extends through a tube wall opening in the containment shell **50**. A strain relief **82** protects the high voltage insulated wire **80** from contact with the edges of the tube opening. Shown in FIG. **6**, the high voltage insulated wire **80** extends from the tube wall opening to the cylindrical inner electrode **56** by going around the cylindrical outer electrode **54** and dielectric sleeve **52**.

[0052] In one embodiment, the inner electrode **56** has a cylindrical shape and includes an opening or cavity which extends from one end of the cylinder to the opposite end whereby air may freely pass through the opening or cavity without obstruction. The outer electrode **54** has a larger diameter than, and is coaxially aligned with, the inner electrode **56**. Air may also flow between the containment shell **50** and the outer electrode **54** as it passes longitudinally through the shell.

[0053] When a high voltage differential is placed between the outer electrode **54** and the inner electrode **56**, oxygen passing inside the containment shell **50** is converted to ozone by combining a single oxygen atom with an oxygen molecule to form ozone. The containment shell **50** is conductive and preferably grounded to the vehicle. Alternately, the containment shell may be non-conductive.

[0054] The ozone generator is powered by connection to the variable high voltage and adjustable frequency power supply. A DC to AC inverter converts the 12 VDC battery voltage to at least about 3000 volts AC, and as high as about 32,000 volts AC or more, at a frequency preferably between

about 300 and 3,000 Hertz. Both the output voltage and frequency are adjustable. The metal electrodes are attached to the high voltage output of the inverter which in turn produces ozone from the oxygen in the air passing within the shell. In order to achieve maximum efficiency in fuel savings and smog reduction, the entire screen tube is preferably installed directly between the existing air filter and the turbo charger of the diesel engine. The incoming filtered clean air is passed through the electrically charged electrode **5** plates, where the oxygen in the air is converted to ozone and subsequently enters the combustion chamber of the diesel engine. The ozone facilitates the diesel fuel to burn more efficiently and minimizes the residual exhaust particulars, wherein the output of the ozone generator **10** is fed through a generator filter where it combines with the air flowing through the air filter **40, 90** and turbo charger fan **96** and feeds to the intake of an internal combustion engine.

[0055] FIG. **7A** is a block diagram of a second embodiment of an ozone generation system for a diesel engine according to the present invention;

[0056] FIG. **7B** is a block diagram of a third embodiment of an ozone generation system for a diesel engine according to the present invention;

[0057] FIG. **8** is a block diagram of a high voltage circuit for supplying power to the ozone generator. An electronic circuit **15** powers the ozone generator **10** and includes a 12-volt battery **20** and a power inverter **18** having a high ratio step-up transformer **22**. The circuit **15** includes a power detector **24** which monitors the voltage from the power inverter **18** and sends power to illuminate a lamp **26** when voltage is detected. One terminal **20** of the power inverter **18** may be connected to a ground terminal **28**. The voltage is then fed to the ozone generator **10**. FIG. **9** is a block diagram of another embodiment of the high voltage circuit **15'**. The high voltage circuit **15'** includes the power detector **24** at the low voltage side of the circuit rather than the high voltage side of the circuit **15'**.

[0058] FIG. **10** is a schematic diagram **110** of the components for the power inverter **18** shown in FIGS. **8** and **9**. The feedback information may control the voltage output, frequency output, or voltage and frequency of the power supply in order to control the amount of ozone delivered to the internal combustion engine. As the speed or power output of the engine increases, the power circuit self-adjusts to provide more ozone to the engine since more oxidation of the fuel is necessary. Inversely, as the speed of the engine or the power output decreases, the feedback information is sent to the power supply in order to reduce the generation of ozone. This will prevent over-generation of ozone. This voltage, by magnetic action through transformer **T1**, steps up the 12 volts direct current to as high as 32,000 volts. The output voltage is adjustable from about 3,000 volts to 32,000 volts and the frequency is adjustable. The circuit preferably provides a high voltage alternating semi-square wave to the ozone generator.

[0059] The circuit **110** uses a pair of transistors **Q3, Q4** in a push-pull configuration to produce a voltage across the primary coil of a transformer **T1** having a center tap connected to **35** the 12v supply positive terminal. The transformer **T1** secondary voltage is applied across the ozone generator terminals and the secondary center tap of the transformer **T1** is connected to the ground terminal. This circuit configuration with the center taps of the primary and secondary coils of transformer **T1** connected to the terminals of the 12-volt voltage source reduces stress on the transformer coils, allowing for a smaller transformer size compared to a circuit which does not implement the transformer center tap in this way.

[0060] FIG. **11** is a block diagram showing a system **200** for controlling production of ozone for an internal combustion engine. The system **200** includes an ozone generator **210** mounted to an engine **250** of a vehicle. As shown in greater detail in FIGS. **3-6**, the ozone generator includes a tubular containment shell **50** having a first and second opening for allowing air to flow therethrough and first and second cylindrical electrodes **54, 56** disposed inside the tubular containment shell **50** permitting air to flow through. The system **200** includes a high voltage power inverter **230** connected with a wire **270** between the first and second cylindrical electrodes **54, 56**, of the ozone generator; the high voltage power inverter **230** capable of producing a voltage differential between

the first and second electrodes **54, 56** sufficient to cause the production of ozone. The ozone generator **210** is disposable in an air intake system of a vehicle having the internal combustion engine, the ozone generator permitting air to flow therethrough wherein a change from oxygen to ozone occurs in the air flowing through the ozone generator and into the internal combustion engine. The system **200** includes at least one control module **220** for controlling production of ozone by the ozone generator **210**. The control module **220** communicates with the power inverter **230**, the ozone generator **210**, a sensor **240** on the ozone generator **210** and the internal combustion engine **250**. The control module **220** may be a frequency control module or a voltage control module for controlling the ozone generator voltage for maximizing performance. The control module **220** may be a temperature control module for activating a cooling system of the ozone generator or the control circuit. The control module **220** may be a flow control module for adjusting air flow through the ozone generator. The control module **220** may be a timer control module for determining state of ozone generator (ON/OFF) as a function of mileage and time and for keeping track of mileage and time. The control module **220** may be an emergency shutoff module.

[0061] In one embodiment shown in FIG. **12**, the system **202** includes at least one monitor module **280**. The monitor module **280** communicates with the at least one control module **220**, a sensor **240** on the ozone generator **210**, and the internal combustion engine **250**. The monitor module **280** includes monitoring features which may be incorporated into the at least one control module. The monitor module **280** detects engine parameters and monitors the production of ozone. The monitor module **280** may be a frequency monitor module or a voltage monitor module having a voltage sensor **240** disposed on the ozone generator for determining the voltage supplied to the ozone generator **210**. The monitor module **280** may be a temperature monitor module including at least one temperature sensor for determining the temperature of the ozone generator, control circuit or a portion of the internal combustion engine. The monitor module **280** may be an air flow monitor which monitors air flow through the ozone generator. The monitor module **280** may be a timer monitor module for generating time interval for the control circuit to work with. The monitor module **280** may be an emergency monitor module for alerting a user to an engine or generation system having an out of range parameter. The monitor module monitors structural parameters including the integrity of each component of the ozone generator. The monitor module monitors fuel parameters including instant fuel usage, average fuel usage, fuel put into the fuel storage tank, and other static and/or dynamic fuel functions. The at least one monitor module **280** may be integrated with the at least one control module **220** to perform both monitoring and controlling functions.

[0062] FIG. **13** is a diagram showing the system **900** for maintaining an internal combustion engine. The system **900** uses ozone for cleaning or maintenance of the internal combustion engine. FIG. **14** is a diagram showing safety sensors **340** for sensing ozone outside of the containment shell. The safety sensors **340** are preferably an ozone detection cell **340a** for sensing ozone outside of the containment shell. Voltage sensors **340b**, pressure sensors **340c** or flow sensors **340d** may be used inside or on the outside of the containment shell **50**. The system **900** includes a case **910** having a case housing and an inner cavity. The case **910** includes a user control panel **920** having a plurality of user controls **922**, the user control panel **920** disposed on the case housing. The case **910** includes an ozone generator **950** in the inner cavity. As shown in FIG. **19** the ozone generator **950** including a containment shell **50** having a first and second opening for allowing air to flow therethrough, a first and second concentric cylindrical electrode **54,56** disposed inside the containment shell **50** and a controllable high voltage source **350** connected between the first and second cylindrical electrode **54,56** wherein the containment shell **50** permits air to flow therethrough wherein a change from oxygen to ozone occurs in the air flowing through the containment shell **50**. The case **910** includes a blower **940** having a first blower end for intake of air external to the case **910** and a second blower end connected to the containment shell first opening.

The system includes a safety module **935** connected with a transmitting medium or wire **924** to the user control panel **920**. The system **900** includes at least one safety sensor **340** connected with a transmitting medium **926** to the user control panel **920** or to the safety module **935**. The transmitting medium **926** may be a wire or wireless signal. The system **900** includes a control module **930** connected with a transmitting medium or wire **924** to the user control panel **920** and the safety module **935**, the control module **930** for controlling voltage applied through wires **923**, between the first and second concentric cylindrical electrode **54,56**. The control module **930** is connected to the user control panel **920** through transmitting medium **925**. The transmitting medium **925** may be a wire or wireless signal. The system **900** includes a tube connector **960** having a first tube connector end connected to the containment shell second opening and a second connector end **970** disposed on the case housing **910**. The system **900** includes an ozone transport tube **980** having a first tube end **978** scalable to the second tube connector end **970** and a second tube end **990** for connection to the internal combustion engine where ozone is delivered to the engine air intake. The blower **940** directs the ozone through the tube ozone generator **950**, the tube connector **960** and the ozone transport tube **980** during operation of the internal combustion engine, whereby the ozone reacts with deposits in the internal combustion engine for removal of the deposits. Controlling the voltage applied between the first and second concentric cylindrical electrode may include controlling amplitude of the voltage or controlling frequency of the voltage or controlling the amplitude and frequency of the voltage. The system **900** may include a dielectric element disposed between the first and second cylindrical electrode. The system **900** may include an engine monitor module for detecting engine parameters.

[0063] FIG. **15** is a flowchart **400** showing a method for maintaining an internal combustion engine. The method includes a step **410** of providing a technician a maintenance system such as the system shown in FIG. **13**. The method includes a step **410** of connecting the second tube end of the ozone transport tube to an air intake of the internal combustion engine. The method includes a step **420** of ensuring the internal combustion engine is running. The method includes a step **430** of commencing production of the ozone and directing the ozone through the ozone transport tube during operation of the internal combustion engine. The method includes a step **440** of allowing the ozone to react with deposits in the internal combustion engine for a specific timing interval or specific amount of time. The method includes the a step **450** of disconnecting the second tube end of the ozone transport tube from the air intake of the internal combustion engine. The method may include a step **460** of measuring results and a step **470** of providing a customer a customer with the results to improve probability of a follow up visit, which should be requested at the time of current service.

[0064] FIG. **16** shows a breakdown **500** of the step **430** of commencing production of the ozone and directing the ozone through the ozone transport tube during operation of the internal combustion engine and the step **440** of allowing the ozone to react with deposits in the internal combustion engine for a specific timing interval or specific amount of time. Safety features are verified **510** and the control circuit sends **520** high voltage to the ozone generator. The control module adjusts **530** air flow through the ozone generator. The control module includes a timer control which determines **550** an amount of time ozone is generated by the ozone generator. The control module includes an emergency shutoff for terminating **560** production of the ozone. The control module then provides **570** the user an audio of visual alarm to indicate termination. The control module monitors structural parameters including integrity of each component of the ozone generator.

[0065] FIG. **17** is a flowchart **600** showing selection of voltage applied to the ozone generator including a first step **610** where feedback information controls **620** the voltage output, controls **630** frequency output, or controls **620**, **630** voltage and frequency of the power supply in order to control the amount of ozone delivered to the internal combustion engine. As the speed or power output of the engine increases **640**, the power circuit self-adjusts **650** to provide more ozone to the

engine since more oxidation of the fuel is necessary. Inversely, as the speed of the engine or the power output decreases, the feedback information is sent to the power supply in order to reduce the generation of ozone. This will prevent over generation of ozone.

[0066] FIG. **18** is a flowchart **700** showing a method of using ozone for maintaining an internal combustion engine. The method uses ozone for maintaining an internal combustion engine using the system described above. The method includes a step **710** of connecting the containment shell to an air intake of the internal combustion engine, a step **720** of ensuring the internal combustion engine is running and a step **730** of commencing production of the ozone and a step **740** of directing ozone into the air intake of the internal combustion engine. The method **700** includes the step **750** of allowing the ozone to react with deposits in the internal combustion engine including timing feedback of providing timing intervals of ozone production based on engine performance feedback.

[0067] FIG. **19** is a flowchart **800** showing a method of using ozone for maintaining an internal combustion engine. The method uses ozone for maintaining an internal combustion engine using the system described above. The method includes a step **810** of connecting the containment shell to an air intake of the internal combustion engine, a step **830** of commencing production of the ozone and a step **840** of directing ozone into the air intake of the internal combustion engine whereby the ozone reacts with deposits in the internal combustion engine. The method includes providing timing intervals of ozone production based on AI feedback through a AI database system trained on vehicle manufacturing data and system feedback. The step **850** of terminating production of ozone is based on the system making the decision that the engine maintenance should be terminated.

[0068] In some embodiments the method or methods described above may be executed or carried out by a computing system including a tangible computer-readable storage medium, also described herein as a storage machine, that holds machine-readable instructions executable by a logic machine (i.e. a processor or programmable control device) to provide, implement, perform, and/or enact the above-described methods, processes, and/or tasks. When such methods and processes are implemented, the state of the storage machine may be changed to hold different data. For example, the storage machine may include memory devices such as various hard disk drives, CD, or DVD devices. The logic machine may execute machine-readable instructions via one or more physical information and/or logic processing devices. For example, the logic machine may be configured to execute instructions to perform tasks for a computer program. The logic machine may include one or more processors to execute the machine-readable instructions. The computing system may include a display subsystem to display a graphical user interface (GUI) or any visual element of the methods or processes described above. For example, the display subsystem, storage machine, and logic machine may be integrated such that the above method may be executed while visual elements of the disclosed system and/or method are displayed on a display screen for user consumption. The computing system may include an input subsystem that receives user input. The input subsystem may be configured to connect to and receive input from devices such as a mouse, keyboard, or gaming controller. For example, a user input may indicate a request that certain task is to be executed by the computing system, such as requesting the computing system to display any of the above-described information, or requesting that the user input updates or modifies existing stored information for processing. A communication subsystem may allow the methods described above to be executed or provided over a computer network. For example, the communication subsystem may be configured to enable the computing system to communicate with a plurality of personal computing devices. The communication subsystem may include wired and/or wireless communication devices to facilitate networked communication. The described methods or processes may be executed, provided, or implemented for a user or one or more computing devices via a computer-program product such as via an application programming interface (API).

[0069] Since many modifications, variations, and changes in detail can be made to the described embodiments of the invention, it is intended that all matters in the foregoing description and shown

in the accompanying drawings be interpreted as illustrative and not in a limiting sense. Furthermore, it is understood that any of the features presented in the embodiments may be integrated into any of the other embodiments unless explicitly stated otherwise. The scope of the invention should be determined by the appended claims and their legal equivalents.

[0070] In addition, the present invention has been described with reference to embodiments, it should be noted and understood that various modifications and variations can be crafted by those skilled in the art without departing from the scope and spirit of the invention. Accordingly, the foregoing disclosure should be interpreted as illustrative only and is not to be interpreted in a limiting sense. Further it is intended that any other embodiments of the present invention that result from any changes in application or method of use or operation, method of manufacture, shape, size, or materials which are not specified within the detailed written description or illustrations contained herein are considered within the scope of the present invention.

[0071] Insofar as the description above and the accompanying drawings disclose any additional subject matter that is not within the scope of the claims below, the inventions are not dedicated to the public and the right to file one or more applications to claim such additional inventions is reserved.

[0072] Although narrow claims are presented herein, it should be recognized that the scope of this invention is much broader than presented by the claim. It is intended that broader claims will be submitted in an application that claims the benefit of priority from this application.

[0073] While this invention has been described with respect to at least one embodiment, the present invention can be further modified within the spirit and scope of this disclosure. This application is therefore intended to cover any variations, uses, or adaptations of the invention using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this invention pertains and which fall within the limits of the appended claims.

[0074] Thus, having described the invention, what is claimed is:

Claims

1. A system for using ozone for an internal combustion engine, the system comprising: an ozone generator comprising: a containment shell having a first and second opening for allowing air to flow therethrough; first and second concentric cylindrical electrodes disposed inside the containment shell; and a high voltage source connected between the first and second concentric cylindrical electrode, the high voltage source capable of producing a voltage differential between the first and second concentric cylindrical electrode sufficient to cause production of ozone; wherein the ozone generator permits air to flow therethrough wherein a change from oxygen to ozone occurs in the air flowing through the ozone generator; at least one safety module including at least one ozone sensor for sensing the ozone exterior to the containment shell; and a control module for controlling the voltage differential from the high voltage source; wherein the control module includes an input from the at least one ozone sensor.
2. The system according to claim 1 wherein controlling the voltage includes controlling amplitude of the voltage or controlling frequency of the voltage or a combination of controlling the frequency and the voltage.
3. A method of using the system according to claim 1 for maintaining an internal combustion engine, the method comprising: providing the system according to claim 1; connecting the containment shell to an air intake of the internal combustion engine; ensuring the internal combustion engine is running; commencing production of the ozone and directing ozone into the air intake of the internal combustion engine; allowing the ozone to react with deposits in the internal combustion engine; providing timing intervals of ozone production based on sensor feedback;

4. A method of using the system according to claim 1 for maintaining an internal combustion engine, the method comprising: providing the system according to claim 1; connecting the containment shell to an air intake of the internal combustion engine; ensuring the internal combustion engine is running; commencing production of the ozone and directing ozone into the air intake of the internal combustion engine; allowing the ozone to react with deposits in the internal combustion engine; providing timing intervals of ozone production based on engine performance feedback;
5. A method of using the system according to claim 1 for maintaining an internal combustion engine, the method comprising: providing the system according to claim 1; connecting the containment shell to an air intake of the internal combustion engine; ensuring the internal combustion engine is running; commencing production of the ozone and directing ozone into the air intake of the internal combustion engine; allowing the ozone to react with deposits in the internal combustion engine; providing timing intervals of ozone production based on AI feedback through a AI database system trained on vehicle manufacturing data and system feedback.
6. A system for using ozone for cleaning of an internal combustion engine, the system comprising: a case having a case housing and an inner cavity, the case including: a user control panel having a plurality of user controls, the user control panel disposed on the case housing; an ozone generator in the inner cavity, the ozone generator including a containment shell having a first and second opening for allowing air to flow therethrough, a first and second concentric cylindrical electrode disposed inside the containment shell and a controllable high voltage source connected between the first and second cylindrical electrode wherein the containment shell permits air to flow therethrough wherein a change from oxygen to ozone occurs in the air flowing through the containment shell; a blower having a first blower end for intake of air external to the case and a second blower end connected to the containment shell first opening; a safety module including at least one safety sensor connected thereto; a control module connected to the user control panel and the safety module, the control module for controlling voltage applied between the first and second concentric cylindrical electrode; and a tube connector having a first connector end connected to the containment shell second opening and a second connector end disposed on the case housing; and an ozone transport tube having a first tube end sealable to the second tube connector end and a second tube end for connection to the internal combustion engine; wherein the blower directs the ozone through the tube ozone generator, the tube connector and the ozone transport tube during operation of the internal combustion engine, whereby the ozone reacts with deposits in the internal combustion engine for removal of the deposits.
7. The system according to claim 6 wherein controlling the voltage applied between the first and second concentric cylindrical electrode includes controlling amplitude of the voltage or controlling frequency of the voltage or controlling the amplitude and frequency of the voltage.
8. The system according to claim 6 including a dielectric element disposed between the first and second cylindrical electrode.
9. The system according to claim 6 including an engine monitor module for detecting engine parameters.
10. A method of using ozone for maintaining an internal combustion engine, the method comprising: providing a maintenance system comprising: a case having a case housing and an inner cavity, the case including: a user control panel inside the inner cavity; an ozone generator inside the inner cavity; a blower inside the inner cavity, the blower having a first blower end for intake of air external to the case and a second blower end connected to the ozone generator; and a safety module including at least one safety sensor connected thereto; a control module connected to the ozone generator and the safety module, the control module for controlling a voltage applied to the ozone generator; a tube connector having a first connector end connected to the ozone generator and a second connector end disposed on the case housing; an ozone transport tube having a first tube end sealable to the second tube connector end and a second tube end for connection to the internal

combustion engine; connecting the second tube end of the ozone transport tube to an air intake of the internal combustion engine; ensuring the internal combustion engine is running; commencing production of the ozone and directing the ozone through the ozone transport tube during operation of the internal combustion engine; allowing the ozone to react with deposits in the internal combustion engine; stopping production of the ozone; and disconnecting the second tube end of the ozone transport tube from the air intake of the internal combustion engine.

11. The method according to claim 10 wherein the control module adjusts air flow through the ozone generator.

12. The method according to claim 10 wherein the control module includes a timer control for determining an amount of time ozone is generated by the ozone generator.

13. The method according to claim 10 wherein the control module includes an emergency shutoff for terminating production of the ozone.

14. The method according to claim 10 wherein the control module monitors structural parameters including integrity of each component of the ozone generator.
